Online references

RCF9112 - Up-to-date RFC for HTTP/1.1. It obsoletes previous RFC729x family.
webserv: Building a Non-Blocking Web Server in C++98 - A Medium article on webserv.
cclaude42 webserv repo - An implementation of webserv made by cclaude42.
What is HTTP? Protocol Overview for Beginners - A freeCodeCamp article on HTTP protocol.
An overview of HTTP - mdn web docs reference on HTTP.
HTTP: The Fundamentals - Dev.to Article on HTTP protocol.

Learning HTTP/1.1: A Comprehensive Guide

HTTP/1.1 is the foundation of the web, so understanding it is crucial for anyone involved in web development, network engineering, or system administration.

Understanding the Basics

- Start with the RFCs: The official specifications for HTTP/1.1 are found in RFC 9112. While it might be dense, it's the ultimate authority.
- Learn the Request-Response Cycle: Grasp the fundamental concepts of how clients and servers communicate.
- Master HTTP Methods: Understand the different HTTP methods (GET, POST, PUT, DELETE, etc.) and their uses.
- HTTP Headers: Learn about the various headers used for request and response, such as Content-Type, User-Agent, and cookies.
- Status Codes: Familiarize yourself with common HTTP status codes (200, 404, 500, etc.) and their meanings.

Practical Application

- Use Developer Tools: Most modern browsers have developer tools that allow you to inspect HTTP requests and responses. This is a great way to see HTTP in action.
- Build Web Applications: Create simple web applications to solidify your understanding of HTTP concepts.
- Analyze Network Traffic: Use tools like Wireshark to capture and examine HTTP traffic.
- Understand HTTP/1.1 Limitations: Learn about the challenges of HTTP/1.1, such as head-of-line blocking and connection management. This will help you appreciate the improvements in HTTP/2 and HTTP/3.

Recommended Resources

- MDN Web Docs: Provides a clear and concise overview of HTTP.
- Online Tutorials: There are many online tutorials and courses that cover HTTP in detail.
- Books: Several books on web development or network engineering have chapters dedicated to HTTP.

Deep Dive Topics

- HTTP Caching: Learn how to optimize website performance using HTTP caching mechanisms.
- HTTP Authentication: Understand different authentication methods (basic, digest, OAuth).
- HTTP Pipelining and Keep-Alive: Explore techniques for improving HTTP performance.
- HTTP/2 and HTTP/3: Learn about the evolution of HTTP and the benefits of newer versions.